

Element E

Soft Principles

Cooperation

One of the biggest STEM principles we applied during this process was cooperation and communication between our three teams. Throughout the process, this looked like having team meetings, sending people to work at other teams temporarily, and having shared documentation between teams such as CAD files and Google Docs.

Budgeting

At the start of the project, we were given a budget of \$250 per group (\$750 in total) to build our solution. We went through our initial design process using this budget, which influenced our choices in materials, complexity, and overall durability. However, our budget has continued to shrink throughout the project due to extenuating circumstances, to the point where we have about \$300 in total to work with. This has forced us to adapt and rethink our solution more than once.

Technical Principles

Material Selection

While selecting the type of plastic we would use for our floors and walls, we evaluated a number of criteria:

Mechanical Properties

One of the biggest requirements of our plastic choice will be to hold up to the dynamic and static forces that will result of housing a komodo dragon. Things we will take into consideration include are the materials’:

- Elastic modulus
- Yield strength
- Density

Water Resistance

To help keep our box clean, we are looking for a plastic that resists water well. This can be identified through metrics like water absorption percentage, polarity, and hydrophobicity.

Stress/Impulse Analysis

For the most part, our box will have to deal with two main types of stress:

- 1) Supporting the dragon’s weight when lifted
- 2) Resisting any thrashing from the dragon, specifically in the tail or head areas

Static Load Analysis - Polypropylene Floor

The floor of our enclosure is made of **¼ inch (6.35 mm) thick polypropylene sheet**. To determine if it can support the dragon's weight, we modeled the floor panel as a **flat plate supported at its edges** (resting on the steel frame), subjected to a **point load** in the center (worst-case scenario).

For this calculation, we know that:

- The maximum weight to be supported is less than 100 lbs (45 kg)
- The load due to this weight is $F = mg = 45 \cdot 9.81 = 441.45 \text{ N}$
- The floor spans a rectangular section, with a length (L) of 36 in (0.9144 m) and a width (W) of 20 in (0.508 m)
- Simply supported on all edges
- Polypropylene mechanical properties:
 - Elastic modulus (E) = 1.5 GPa
 - Yield strength(σ) = 30 MPa
- Poisson's ratio (ν) = 0.4

Using the formula for maximum deflection of a rectangular plate under a center point load:

$$\delta_{max} = \frac{F a^2 b^2}{\pi^6 D (a^2 + b^2)}$$

We must first calculate the flexural rigidity (D), using the formula:

$$D = \frac{Et^3}{12(1-\nu^2)} = \frac{1.5 \cdot 10^9 \cdot (.00635)^3}{12(1-.04^2)} = 32.057$$

Now we can find the maximum deflection to be:

$$\frac{441.45 \cdot \left(\frac{0.9144}{2}\right)^2 \left(\frac{.508}{2}\right)^2}{\pi^6 (32.057) \left(\frac{.9144}{2} + \frac{.508}{2}\right)} = 0.0002716$$

Giving us 0.0002716 meters, or .2716 mm, which is well within the bending limits of the polypropylene.

Dynamic Load Analysis - Polypropylene Walls and Steel Frame

The walls of our enclosure are made of **⅛ inch (6.35 mm) thick polypropylene sheet**. To determine if it can withstand the tail thrashing of a dragon, we modeled the floor panel as a **flat plate supported at its edges** (resting on the steel frame), subjected to a **quick impact** from the tail.

For this calculation, we will assume:

- The weight of the tail will be under 150 lbs (68.039 kg)
- The tail length to be 3 feet (0.914 m) long and ½ foot (0.152 m) wide
- The tail will move at a maximum speed of 5 meters per second
- An impact time of .01 s
- The walls spans a rectangular section, with a length (L) of 36 in (0.9144 m) and a width (W) of 20 in (0.508 m)
- Simply supported on all edges
- Polypropylene mechanical properties:
 - Yield strength (σ) = 30 MPa
 - Density (D) = 900 kg/m³

Now we can estimate the impact force using the impulse-momentum theorem to estimate the peak force during the impact:

$$F = \frac{\Delta p}{\Delta t} = \frac{mv}{\Delta t} = \frac{68.039 \cdot 5}{.01} = 34019.5 \quad \text{Giving us a } F = 34,019 \text{ N}$$

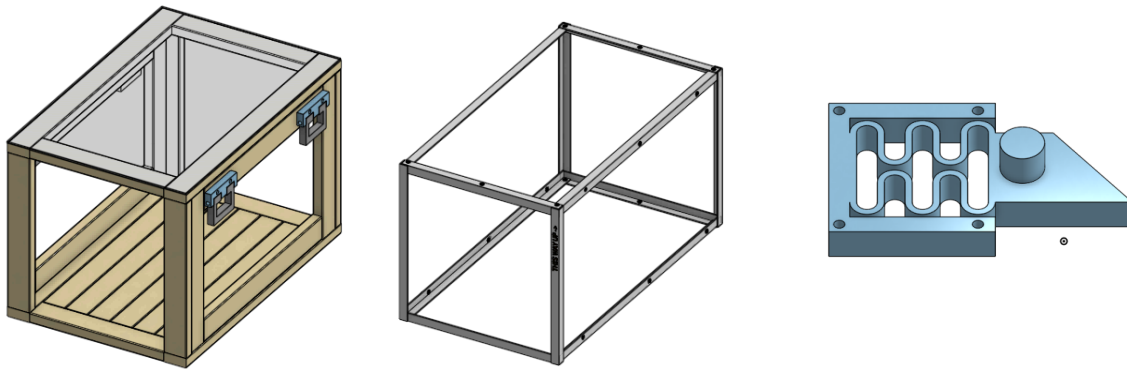
Using this force, we can calculate the stress, using the formula:

$$\sigma = \frac{F}{A} = \frac{34019.5}{.914 \cdot 0.152} = 244871.444201$$

Giving us a pressure of 244,871 Pa, or 0.245 MPa, which is well within the 30 MPa yield strength of our polypropylene.

CAD

One of the key tools we used during this process was a CAD software called OnShape. Our effective use of OnShape allowed us to visualize our designs without having to physically build them every time, which was extremely helpful for figuring out spacing, window design, handle design, creating a bill of materials, and communicating within the team and with experts. We were also able to design a number of custom parts, specifically for the medical windows.



EXPERT REVIEW

A handwritten signature in black ink, appearing to read 'Mr. Kiser'.

Mr. Kiser (Engineering teacher, B.S. in Chemical Engineering)